

A unified genetic and epigenetic model to predict breast cancer intrinsic subtypes using large DNA-level multi-omics data and hierarchical learning

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Abstract— Breast cancer subtyping presents a significant clinical and scientific challenge. The prevalent expression-based Prediction Analysis of the Microarray of 50 genes (PAM50) system and its Immunohistochemistry (IHC) surrogate tests showed substantial inconsistencies and did not apply to the rapidly progressing circulating tumor DNA screenings. We developed Unified Genetic and Epigenetic Subtyping (UGES), a new intrinsic subtype classifier, by integrating large-scale DNA-level omics data with a hierarchy learning algorithm. Our benchmarks showed that both multi-step hierarchical learning and using all DNA-level alteration data are crucial, improving the overall AUC score by over 8.3% compared to the one-step multi-classification method. Based on these insights, we developed UGES a three-step classifier based on 50831 DNA features of 2065 samples, including mutations, copy number aberrations, and methylations. UGES achieved an overall AUC score of 0.963 and greatly improved the clinical stratification of real-world patients, as each subtype strata's survival difference became statistically more significant, $P=9.7e-55$ (UGES) vs. $2.2e-47$ (PAM50). Finally, UGES identified 52 subtype-specific DNA biomarkers that can be targeted in early screening technology to expand the time window for precision care. The UGES code is freely available at <https://github.com/labxscut/UGES>.

Index Terms— Breast cancer subtype classifier; Hierarchical learning; Subtype-specific biomarkers; Multi-omics big-data; Cancer early screening.

I. INTRODUCTION

BREAST cancer is a malignant, complex, and highly heterogeneous tumor that overwhelmingly impacts women. Precise identification of the cancer's subtypes is crucial to optimize treatment and improve outcome for patients [1]. However, breast cancer subtyping remains a

difficult scientific and clinical challenge [2-4]. The most prevalent subtyping system, termed PAM50, can discriminate among four main intrinsic subtypes: Basal-like (**Basal**), Her2-enriched (**Her2**), Luminal A (**LumA**), and Luminal B (**LumB**). This gene expression-based system was first established in the Prediction Analysis of Microarray of 189 patients based on the expression of 50 signature genes (i.e., **PAM50** subtyping) [5]. Clinically, breast cancer intrinsic subtyping was surrogated to Immunohistochemistry tests (**IHC** subtyping) specifically targeting estrogen receptor (ER), progesterone receptor (PR), human epidermal growth factor 2 (Her2), and Ki67 proteins [6, 7].

However, our analysis, along with abundant external evidence, revealed pervasive and significant inconsistencies between PAM50 and IHC subtypes [8-10]. The discrepancies were observed in both the TCGA (U.S.) and METABRIC (U.K.) populations. In the METABRIC dataset, 599 out of 1,086 samples (55.16%) showed discordant PAM50 and IHC subtypes (see **Supplementary Table S1**). Independent studies support these findings. For instance, a study conducted at the Tri-Service General Hospital in Taiwan analyzed 372 breast cancer patients. It found that the IHC subtyping misclassified up to 61.11% of PAM50 Her2-enriched patients as Luminal B [9]. Another study from the Samsung Medical Center ($n=607$) reported classification discrepancies in 38.4% of patients. The misclassification between Luminal-A and Luminal-B types was associated with significantly poorer overall survival ($p<0.001$) [10]. These findings highlighted the dire clinical consequence of such misclassifications.

The inconsistencies between PAM50 and IHC subtyping arise from several factors. First, expression-based subtyping is inherently variable due to the high variability of mRNA expression, which is often influenced by cellular, temporal, and spatial noises [11-13]. As a result, PAM50's classification based on the expression of only 50 genes frequently deviates from the actual intrinsic subtypes, as proven by numerous studies [14, 15]. Second, the IHC surrogate for PAM50 is systematically biased. While using only four biomarkers is convenient in clinical settings, it overlooks rich genome-wide patterns essential for accurate subtyping [16-18]. In fact, intrinsic subtypes are inherently determined by a broader range of genetic and epigenetic factors, i.e., mutations, copy number variations, and methylation alterations [19-21]. Last but not least, both PAM50 and IHC subtyping underestimated inter-tumoral heterogeneity. Developed in the early 2000s with only small sample sizes, the methods missed important

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features that only become identifiable with large-scale omics data [5, 22].

To overcome these challenges and improve subtyping accuracy and consistency, we proposed to identify novel genetic and epigenetic determinants of breast cancer intrinsic subtypes using large-scale data. Based on the identified features, we developed a DNA-level classifier (**Fig. 1a**). Unlike mRNA and protein changes, DNA alterations are more fundamental, discrete, robust, and easier to ascertain [23]. According to the central dogma, genetic information flows from DNA to RNA and then to protein, dictating that DNA-level intrinsic patterns would determine expression and proteomic phenotypes. Additionally, DNA molecules are structurally stable and less prone to transcriptional and translational noises, making their alterations more reliable for subtyping [24]. Notably, early breast cancer detection using the genomic and epigenomic sequencing of circulating tumor DNA, either targeted or genome-wide, is picking up the pace [25]. By integrating these sequencing advances, our DNA-level classifier enables breast cancer subtyping at an early stage, even upon diagnosis, thus significantly extending the time window for precision care.

The integration of multi-omics profiles for cancer subtyping has been widely explored in studies of various cancers [26-28]. DNA alterations, such as mutations [29], copy number aberrations (CNAs) [30, 31], and methylation changes [32, 33], have been used individually or in combination with transcriptomic features for subtyping. For example, Curtis et al. proposed a system of transcriptional processes influenced by methylation and CNAs, demonstrating that CNAs are key drivers of inter-tumor heterogeneity in breast cancer [34]. Islam et al. developed a deep neural network classifier based on expression and CNA data [35], while Lin et al. created a classifier incorporating expression, CNAs, and methylation data [36]. Choi et al. proposed moBRCA-net, which integrates expression, methylation, and miRNA data using attention mechanisms for breast cancer subtyping [37]. Most recently, Rajpal et al. introduced XAI-MethylMarke, a method combining deep learning and explainable AI to identify a set of 52 DNA methylation biomarkers for breast cancer subtype classification [38].

Improving upon these studies, we developed a DNA-level classifier by leveraging the two largest multi-omics datasets – **The Cancer Genome Atlas (TCGA)** and **METABRIC** cohorts ($n=2,065$). This classifier, termed **UGES** (Unified Genetic and Epigenetic Subtyping for Breast Cancer), is shown in **Fig. 1c**. Methodologically, instead of using the conventional one-step multi-class learning approach, we introduced a novel hierarchical learning algorithm designed to reflect the underlying tumor evolution process. Our stepwise hierarchical classification method identifies intrinsic subtypes incrementally, effectively capturing nuanced similarities and differences between subtypes. Compared to the one-step multi-class strategy, our hierarchical method better addresses class confusion by optimizing classification tasks at all sublevels. This significantly reduces overall misclassification rates.

This approach proved crucial, improving overall AUC by at least 8.3% compared to the state-of-the-art one-step methods. Additionally, when evaluated against two other competitive hierarchical approaches, the **UGES** algorithm demonstrated superior performance, achieving the best precision-recall performance ($AUC=0.963$) and the highest prognostic prediction power. As an application, we used the UGES-classified subtypes from the **TCGA** and **METABRIC** datasets to conduct a survival analysis. The results demonstrated its clinical relevance, showing significantly improved prediction for patients compared to PAM50. In addition, we implemented an effective Lasso-Logistic model for supervised learning, which soft-selected significant DNA alteration. The follow-up differential analysis identified 52 key DNA alterations that potentially define breast cancer intrinsic subtypes.

II. MATERIALS AND METHODS

A. Data collection and pre-processing

The Cancer Genome Atlas (**TCGA**) [39] and Molecular Taxonomy of Breast Cancer International Consortium (**METABRIC**) [34] datasets, including mutation, CNA, methylation, transcription, and clinical data, were downloaded from the cBioPortal Cancer Genomics Platform (<https://www.cbioportal.org>) [40] and the TCGA data portal (<https://portal.gdc.cancer.gov/>).

We summarized all raw DNA alterations to the gene level. For mutation, we mapped the downloaded mutation profiles to the gene level, resulting in a binary matrix where the i, j -th cell is an indicator of the i -th gene that was mutated at least once in the j -th patient. For CNA, we used the downloaded absolute copy number values, as estimated by the TCGA consortium and mapped to the gene level by cBioPortal. For methylation, we mapped the raw methylation array profiles (HumanMethylation450 BeadChip, Illumina) to the gene level using the *FDdb.InfiniumMethylation.hg19* package in R. In the case of many-to-one mapping between the CpG sites and a gene, we used the average methylation level for the gene.

We intersected DNA features from both datasets, retaining only shared methylation features. DNA alteration features with over 50% missing values were excluded, and missing values in the remaining features were imputed using the 10-nearest neighbor method with the *impute.knn()* function from the *impute* R package [41, 42]. This resulted in the inclusion of 50,831 gene-level DNA features, including 16,770 mutations, 25,594 CNAs, and 8,467 methylation features.

B. Lasso-Logistic regression

We applied the Lasso-Logistic regression model using the *glmnet* package in R [43]) with 10-fold cross-validation for classification. The Least Absolute Shrinkage and Selection Operator (**Lasso**) handles high-dimensional, multi-modal data while avoiding overfitting by regularizing non-informative features to zero [44]. The model's single hyperparameter λ controls the regularization level. To determine the optimal λ value, we used the *cv.glmnet()* function and selected the model with the highest accuracy.

To describe this multinomial model, we used the following annotations. Suppose the response variable has K levels $G = \{1, 2, \dots, K\}$ and each sample x_i in our study has $m = 50,831$ features, i.e., $x_i = \{x_{i,1}, x_{i,2}, \dots, x_{i,m}\}$, then we model:

$$\Pr(G = k|X = x) = \frac{e^{\beta_{0k} + \beta_k^T x}}{\sum_{l=1}^K e^{\beta_{0l} + \beta_l^T x}} \quad (1)$$

Let Y be the $N \times K$ indicator response matrix, with elements $y_{il} = I(g_i = l)$. Then the Lasso elastic net penalized negative log-likelihood function becomes:

$$l(\{\beta_{0k}, \beta_k\}_1^K) = -\left[\frac{1}{N} \sum_{i=1}^N \left(\sum_{k=1}^K y_{ik} (\beta_{0k} + x_i^T \beta_k) - \log \left(\sum_{l=1}^K e^{\beta_{0l} + x_i^T \beta_l} \right) \right) \right] + \lambda \sum_{j=1}^p |\beta_j| \quad (2)$$

where β is a $p \times K$ matrix of coefficients, β_k is the k -th column (for outcome category k), β_j the j -th row (vector of K coefficients for variable j), λ is the penalty parameter, and N is the total number of samples.

C. Hierarchical learning algorithm

We extended the hierarchical learning algorithm proposed by Bengio et al. [45] to construct the optimal multi-step classification training framework (see **Algorithm 1**).

Algorithm 1: the UGES training framework

Input:

$X_{train} = (X_1, X_2 \dots X_n)$ & $Y_{train} = (Y_1, Y_2 \dots Y_n)$; $k = 4$;
The subtype hierarchy: $T = \{\}$.

Repeat until $k=1$:

1. Train by cross-validation k One-vs-Rest classifiers

$\bar{f}_1^{(k)}, \dots, \bar{f}_k^{(k)}$ using Lasso-Logistic regression.

2. Compute the pairwise classification error, the confusion matrix $\bar{C}_{ij}$ and the symmetrized confusion matrix A :

$$\begin{aligned} \text{i. } \bar{C}_{ij} &= \left| \left\{ (x, y_i) \in X_{test} : \arg\max_r \bar{f}_r^{(k)}(x) = j \right\} \right| \\ \text{ii. } A &= \frac{1}{2} (\bar{C} + \bar{C}^T) \end{aligned}$$

3. Merge the two most similar subtypes to an internal subtype $c = a \cup b$ s.t. $(a, b) = \arg\max_{(i,j)} A_{ij}$ and add c to T .

4. $k=k-1$; update all sample labels of a and b to c in Y .

Output:

The learned subtype hierarchy T and its associated multi-step hierarchical classifiers $\bar{f}_i^{(\cdot)}$.

The training framework is designed to build hierarchical classifiers by iteratively merging subtypes based on their classification similarity. The process begins with the initialization of an empty subtype hierarchy, denoted as T . In each iteration, a set of k one-vs-rest classifiers is trained using Lasso-Logistic regression, followed by cross-validated to assess classifier performance. The pairwise classification error between subtypes is calculated to generate a confusion matrix that captures the frequency of misclassifications between subtype pairs. The confusion matrix is symmetrized to account for reciprocal errors, providing a standardized metric for measuring subtype similarity.

After computing the confusion matrix, the framework identifies the two most similar subtypes based on maximal similarity and merges them into a new internal major subtype. This newly formed major subtype is added to the learned hierarchy, and the sample labels corresponding to the merged subtypes are updated accordingly. This process continues iteratively, merging the most similar subtypes at each step, until only one subtype remains. The final output consists of the learned hierarchical structure of subtypes and the associated multi-step hierarchical classifiers, forming a set of classifiers that operate all sub-levels. This approach also enables learning of the most probably subtype relationships at the same time.

Thus, the framework UGES consists of both the subtype hierarchy T and the set of multi-step hierarchical classifiers f s. In application, UGES guides the decision-making process by navigating through a sequence of hierarchical steps, progressively narrowing down the possible subtypes. Starting from the root of the hierarchy, a sample is passed through a series of classifiers in f s, each of which is trained to distinguish between fewer but more similar subtypes at lower levels of the hierarchy. This multi-step process not only improves classification accuracy by focusing on finer distinctions between closely related subtypes, but also reduces the complexity of the classification task, making it more interpretable and computationally efficient.

D. Relative importance

Genetic and epigenetic determinants of intrinsic subtypes were identified by the Lasso-Logistic regression, enabling the quantification of effects for all three types of DNA alterations. Lasso-Logistic regression allows for high-dimensional variable compression and selection. Consequently, the corresponding regression coefficients of the selected DNA alterations by Lasso were summarized to assess the importance. In total, 236, 367, and 530 DNA alterations were selected from the B-vs-(H,LA,LB), H-vs-(LA,LB), and LA-vs-LB sub-classifiers of UGES, respectively (**Supplementary Table. S2**).

We applied relative importance to measure the total effect of each DNA alteration class [46-48]. A straightforward strategy for estimating the relative importance of a DNA alteration class is to calculate the sum of regression coefficients within that class and express the sum as a percentage. Suppose $f = \{V, C, M\}$ is the union set of all mutation, CNA, and methylation alterations. Then, the relative importance RI_i of alteration i can be calculated by:

$$RI_i = \frac{S_i}{S}, \quad (3)$$

where $S_i = \sum_{i \in f} |c_i|$ is the sum of all absolute regression coefficients for a certain DNA alteration class and $S = \sum_f S_f$.

E. Survival analysis

Univariate and multivariate survival analyses were performed in R using the *survival* and *survminer* packages, with Overall Survival (OS) as the primary endpoint. For the

multivariate analysis, Cox regression was used to assess the effects of UGES subtypes and age with the *coxph()* function. A forest plot was generated to show survival outcomes according to UGES subtypes and age, using a resampled balanced dataset (304 samples for each subtype) as the reference. P-values were adjusted using the Benjamini-Hochberg (BH) method to control for False Discovery Rate, with significance determined based on a BH-corrected threshold of 0.05.

F. Differential analyses

Differential analyses were conducted in R on a set of 112 DNA alteration features, including the top 10% of alterations selected by sub-classifier. For mutation data, the Chi-Square Test was applied using the *chisq.test()* function. For copy number aberration data, the Kruskal-Wallis Test was performed using the *kruskal.test()* function. For methylation data, the Welch ANOVA Test was performed using the *oneway.test()* function, with parameter *var.equal = FALSE*. Features were considered significantly different if the BH-corrected test p-value was < 0.01 . All the statistical tests were two-sided.

III. RESULTS

A. Cohort characteristics and data separation

We downloaded and analyzed DNA-level and clinical data for a total of 2065 samples, including 931 TCGA samples and 1134 METABRIC samples (**Fig. 1b**). We assigned these samples to three breast cancer cohorts: TCGA-only cohort, METABRIC-only cohort, and combined TCGA+METABRIC cohort. The descriptive statistics of demographics (age) and relevant clinical variables (ER status, PR status, Her2 status) were collected for all cohorts (see **Table. 1**).

Univariate survival analyses of clinical variables also yielded confirmatory statistics suggesting that all the cohorts represented the general breast cancer population and were suitable for integrative analysis. Elderly patients had a statistically significant worse survival rate than expected [49, 50], with a relative increase of death risk of approximately 4% ($HR = 1.039$, 95% $CI = [1.029, 1.049]$, $P < 2e-16$), demonstrating that the clinical variables were generally consistent across the cohorts. Patients with positive ER status or negative Her2 status also had a statistically significantly improved survival rate than expected [51, 52], with a relative reduction of death risk by $>25\%$ or $>64\%$ (positive ER status: $HR = 0.749$, 95% $CI = [0.563, 0.997]$, $P = 0.0474$; positive Her2 status: $HR = 1.643$, 95% $CI = [1.244, 2.169]$, $P = 0.0005$).

We divided the TCGA+METABRIC cohorts into training and test sets (**Fig. 1c**). To ensure balanced representation of the four intrinsic subtypes in the training set, we randomly selected 200 samples from each subtype, resulting in a total of 800 samples for training, with the remaining 1,265 samples used as the test set. We trained and cross-validated DNA-level classifiers based on all strategies with the training set and then applied the models to the test set. We used Receiver Operating

Characteristic (ROC) curves and Area Under Curve (AUC) scores as performance evaluation metrics. For technical details of data processing, see **Materials and Methods**.

B. UGES -- a sufficient and accurate DNA-level classifier

We first developed a sufficient yet trimmed DNA-level classifier for distinguishing intrinsic subtypes. One drawback of multi-omics subtyping is its mixed and undifferentiated molecular features, including genomic, epigenomic, and transcriptomic data. In contrast, a restrained yet effective DNA-level classifier can improve model interpretability, applicability, and robustness. To achieve this, it is essential to determine the optimal classification strategy and the specific DNA-level features to include. Both the TCGA and METABRIC datasets provide comprehensive DNA-level alterations, including mutations, CNAs, and methylations. Before using the complete set of DNA-level alterations to evaluate candidate hierarchical classification, we assessed the informativeness of individual DNA-level alteration categories in combination with four potential strategies.

The first candidate strategy was a three-step hierarchical classification approach derived from the results of the proposed hierarchical learning algorithm (see **Algorithm 1** in **Materials and Methods**). The approach was encoded as **(B,(H,(LA,LB)))** by applying the standard *Newick* tree annotation (**Fig. 2a**) [53], where **B**, **H**, **LA**, and **LB** denote the Basal, Her2, Lum-A, and -B subtypes, respectively. In this strategy, the Basal subtype was separated from all other subtypes at the first step. Then, Her2 was separated at the second step. Finally, Luminal-A and Luminal-B were distinguished in the last step.

We also applied three other classification strategies (**Fig. 2b**). The first is the most common simple one-step multi-class classification [54], denoted as **(B,H,LA,LB)**, which classifies four intrinsic subtypes all at once. The second strategy, **(B,H,(LA,LB))**, is a two-step classification inspired by the clinical decision process. In this approach, any given patient was first tested for the Basal and Her2 biomarkers, followed by LumA and LumB sub-markers for further subtyping [54, 55]. This strategy is reasonable, as the gene expression profile of Luminal-like subtypes are considered more like each other compared to other intrinsic subtypes [56, 57]. Finally, to address the underrepresentation of Basal and Her2 subtypes in the cohorts, we introduced a new strategy **((B,H),(LA,LB))**. In this approach, Basal and Her2 subtypes are grouped at the first classification step to boost their sample size, thereby balancing the training set. This adjustment improves learning efficiency and model generalizability [58].

Overall, we found that the self-learned **(B,(H,(LA,LB)))** model using our hierarchical learning algorithm, outperformed the other strategies. Indeed, all multi-step strategies had higher performance compared to the one-step approach, highlighting the importance of incorporating hierarchical classification into the task. As shown in **Fig. 2c**, the training and testing overall AUC scores (averaged over all subtypes) for **(B,(H,(LA,LB)))** were 0.984 and 0.963, respectively. In comparison, the scores

for ((B,H),(LA,LB)) were 0.991 and 0.952, for (B,H,(LA,LB)) were 0.984 and 0.956, and for (B,H,LA,LB), they were only 0.965 and 0.889. The fact that all AUC scores for these strategies were greater than 0.9 supports our conclusion that DNA-level information is sufficient and robust for identifying intrinsic subtypes accurately.

Notably, all multi-step strategies grouped Luminal-like subtypes at the initial step, resulting in a substantial improvement in the overall AUC by at least 6.3% compared to the one-step strategy model, which was underlying PAM50 and many other multi-omics subtyping methods. This result gave strong empirical evidence of the common genetic and epigenetic basis of Luminal-like tumors as they diverged further from Basal and Her2 subtypes. Such commonality was completely not exploited in one-step multi-class subtyping. We also compared the three strategies using the ROC curves of the combined set by the one-vs-one method (**Supplementary Fig. S1**). It was obvious that (B,H,LA,LB) was the worst at distinguishing LumA and LumB (AUC=0.828) compared to the multi-step classifiers AUC=0.948 in (B,(H,(LA,LB))), AUC=0.941 in ((B,H),(LA,LB)), AUC=0.951 in (B,H,(LA,LB)). The latter improves the separation accuracy by 11.3% in distinguishing Luminal A and B subtypes, which is the most difficult case.

We found that the (B,(H,(LA,LB))) strategy is the best for identifying Her2 and Luminal-like subtypes. In terms of the test set, its AUC scores are 0.987, 0.974, 0.96, and 0.933 for Basal, Her2, LumA, and LumB subtypes, respectively, while those for ((B,H),(LA,LB)) are 0.991, 0.956, 0.953 and 0.909; for (B,H,(LA,LB)) are 0.992, 0.94, 0.96 and 0.933; and for (B,H,LA,LB) are 0.988, 0.935, 0.852 and 0.782, respectively (**Fig. 2c**). In fact, (B,(H,(LA,LB))) gained performance considerably (AUC +0.018) in identifying the Her2 subtype, and performed best in identifying LumA and LumB subtypes, with only a slight loss in identifying the Basal cases (AUC - 0.005). Considering the high performance of (B,(H,(LA,LB))), and it was self-learned by the hierarchical learning algorithm, we selected and implemented it into the final UGES classifier.

We also found that multi-omics classifiers using all DNA-level alterations significantly outperformed those relying on a single type of alteration. For example, in the test set, the multi-omics classifier (B,(H,(LA,LB))) achieved an overall AUC of 0.963. However, the AUCs for mutation-only, CNA-only, and methylation-only classifiers were only 0.7, 0.877, and 0.935, respectively, all of which were notably lower. Subtype-wise, the multi-omics classifier (B,(H,(LA,LB))) also consistently outperformed the corresponding single-feature classifiers (**Fig. 2d**). The same pattern was observed for classifiers based on other strategies (**Supplementary Fig. S2**), except for the methylation-only (B,H,LA,LB) classifier, which performed better (AUC=0.943) than the multi-omics classifier (AUC=0.935) in identifying Her2 samples. This exception may be attributed to model overfitting specific to the methylation training data, as discussed later (see **Discussion**). Based on these findings, we decided to include all mutations, CNAs, and methylation features in the final classifier. We finally built UGES as a three-step all-feature (B,(H,(LA,LB))) classifier,

which achieves the highest overall AUC (0.963) in our benchmark analysis.

Additionally, we trained UGES on TCGA and tested it on METABRIC and vice versa. When UGES model was trained solely on METABRIC and tested on TCGA, the overall AUC on the test set was 0.735 (**Supplementary Fig. S3**). In the reverse scenario, when UGES was trained on TCGA and tested on METABRIC, the overall AUC improved to 0.846 (**Supplementary Fig. S3**). These results are acceptable, as training a model on a single cohort typically results in lower performance when tested on an independent cohort. This is primarily due to the batch effect, including differences in patient demographics, sample processing, and sequencing protocols between cohorts. Such differences can introduce domain shifts that limit a model's ability to generalize across datasets while trained solely on a single source. The observed performance decline in both scenarios reflects the inherent limitations of cross-cohort validation, emphasizing the importance of integrative multi-source training. Exposing the model to a broader range of variability and clinical characteristics allows it to address better the challenges posed by dataset heterogeneity.

We further compared UGES with leading methylation and/or transcription profile based methods, such as XAI-MethylMarker [38] and moBRCA-net [37] (**Supplementary Fig. S4**). With the entire test set, which included both TCGA and METABRIC samples, UGES model outperformed XAI-MethylMarker across all metrics, including AUC, accuracy, F1-score, precision, and recall (**Supplementary Fig. S4a**). Specifically, UGES achieved an AUC of 0.963, compared to XAI-MethylMarker's 0.898. Further, to make a fair comparison with moBRCA-net [37], we used only the TCGA part of the test set. UGES outperformed moBRCA-net in all three metrics: accuracy (0.9154 vs. 0.891), weighted F1-score (0.921 vs. 0.887), and Matthews Correlation Coefficient (MCC) (0.8487 vs. 0.831) (**Supplementary Fig. S4b**). These results underscore the effectiveness of the UGES model's DNA-based design, supporting its potential as a robust and effective solution for clinical applications such as liquid biopsy.

C. methylation has the main effect in defining subtypes

We studied the relative importance of each DNA feature based on UGES-defined intrinsic subtypes. Biologically, genetic and epigenetic alterations can have different impacts on intrinsic subtypes via varied biological mechanisms [59, 60], including gene functional disruption, gene dosage changes, and genome accessibility changes. The availability of full DNA-level data on a large scale allowed us to quantify these effects based on the UGES subtyping and to explore subtype-wise early detection and precise treatment options. Relative importance is a metric that calculates the relative contribution of predictive variables (see **Materials and Methods**). The relative importance of DNA features in the UGES sub-classifiers *B-vs-(H,LA,LB)*, *H-vs-(LA,LB)*, and *LA-vs-LB* are shown in **Table 2**.

We found that DNA methylation changes were consistently the most important alteration in defining all sub-classifiers. The relative importance of methylation features was 53.97%, 55.08% and 51.62% for *B-vs-(H,LA,LB)*, *H-vs-(LA,LB)*, and *LA-vs-LB* sub-classifiers, respectively, all over 50%. In general, CNA is a remote second in importance, accounting for 33.35%, 25.15%, and 19.76% of the total importance in the respective models, and mutation is in third place, accounting for 12.68%, 19.78%, and 28.63% of the total in the respective models. Consistent with our quantitative assessment here, the methylation-feature-only classifiers had the highest AUC scores for all subtypes (see **Fig. 2d**), suggesting that methylation change as a whole is probably the most prominent feature in defining the intrinsic status of cancerous cells in breast cancer, which were also reported in previous studies alphabet in smaller cohort sizes [61, 62].

Either CNA or mutation features could be the second most important, depending on the sub-classifier. CNA features had the second largest relative importance in the *B-vs-(H,LA,LB)* and *H-vs-(LA,LB)* sub-classifiers, 33.35% and 25.15%, respectively, but the smallest relative importance in the *LA-vs-LB* sub-classifier, at 19.76%. Mutation features had the second largest relative importance in the *LA-vs-LB* sub-classifier at 28.63% but the smallest in the *B-vs-(H,LA,LB)* and *H-vs-(LA,LB)* sub-classifiers, at 12.68% and 19.78%, respectively. Notably, the relative importance of mutation features was the second highest (28.63%) in the *LA-vs-LB* sub-classifier but the lowest in the other sub-classifiers, suggesting that mutations were crucial in distinguishing LumA and LumB subtypes, as reported by previous research [63].

D. UGES-defined subtypes show improved clinical relevance

We found that UGES subtypes were substantially different from PAM50 subtypes. The alluvial plot and the confusion matrix showed that subtype reassignment occurred for 11.57% (239 in 2065 samples) (**Figs. 3ab**). The reassignment better resolves the classification of LumA and LumB subtypes. Among Basal, Her2, LumA, and LumB PAM50 subtypes, 5.79%, 7.63%, 15.64%, and 9.87% changed subtypes by UGES, with the most frequent change being 11.25% LumA (PAM50) to LumB (UGES). This is plausible because Luminal-like types are genetically and epigenetically more alike than others and are the most challenging to resolve.

UGES increased the model sensitivity to detect the most difficult Her2 intrinsic subtype (+28.92% after reassignment). 94.44% of the Basal (PAM50) subtype changed to Her2 (UGES) subtype (17 out of 18 switchings) and 67.31% of LumB (PAM50) subtype changed to Her2 (UGES) subtype (35 out of 52 switchings), while the frequency of change to other subtypes was much lower, with 0% and 5.56% of Basal (PAM50) subtype changed to LumA and LumB (UGES) subtypes, and 7.69% and 25% of LumB (PAM50) subtype changed to Basal and LumA (UGES) subtypes, respectively.

We then demonstrated that UGES subtypes were more predictive of patients' overall survival as evidence of their validity and clinical relevance. We found that UGES subtypes

showed improved prognostic value with better overall survival (OS) stratification compared to PAM50 subtypes (**Fig. 3cd**). The results were based on both univariate and multivariate survival analyses (see **Materials and Methods**). Based on univariate survival analysis, in five out of six pairwise comparisons (each comparing two subtypes), we found that the survival of UGES subtypes was more distinguishable from each other than PAM50 subtypes (**Fig. 3c**). For example, the statistical significance of survival differences among these UGES subtypes increased (smaller p-values), except between Her2 and LumB subtypes, proving that the UGES subtypes are more clinically relevant. Moreover, under UGES, the survival differences between the LumA and Basal subtypes became statistically significant (p-value decreased from 0.07 to 0.04). In general, UGES subtyping resulted in sharper differences between clinical intrinsic subtype groups.

In multivariate survival analysis, where age was added as a risk factor, UGES subtyping also showed improved clinical relevance. Using a resampled balanced sample set as reference, UGES subtypes were also found to better stratify the intrinsic subtypes for their survival difference ($P = 9.7e-55$), as compared to PAM50 subtypes ($P = 2.2e-47$) (**Fig. 3d**). For example, the significance of the differences between Her2 and the other subtypes had increased, changing from statistically insignificant to significant (p-value decreased from 0.0557 to 0.0048). Moreover, the LumA subtype had a higher overall survival rate (Hazard Ratio < 1) compared to the other subtypes, verifying the better prognosis of LumA patients, as reported by others [64]. These findings proved that UGES subtypes yielded more clinically distinguishable patient subgroups, influential for more precise clinical management.

E. UGES identified signature subtype-specific alterations

Unraveling genetic and epigenetic heterogeneity and identifying subtype-specific markers of breast cancer are key to developing targeted therapies that will lead to improved patient outcomes [65]. Here, we conducted differential analyses using 112 genetic and epigenetic features selected as the top 10% most important features in each UGES sub-classifier (see **Materials and Methods**). As a result, we identified 52 subtype-delineating signature alterations. These signature DNA features are presented in **Table. 3** grouped by their corresponding differential subtypes. Heatmap plots of signature DNA alteration profiles in **Fig. 4** also showed patterns consistent with the results given in **Table. 3**.

The heat patterns are, in general, consistent with the relative importance of DNA alteration classes previously discussed (**Fig. 4**). For example, the B and (H,LA,LB) subtypes differed little in mutation profiles but significantly in CNA and methylation profiles. According to the relative importance analysis, CNA and methylation alterations totally accounted for 87.02% of the feature importance for the *B-vs-(H,LA,LB)* sub-classifier.

We observed four distinct signature heat patterns of respective DNA alteration classes, which are explained as follows: First, gene mutation patterns showed substantial

differences between the subtype groups, with a total of nine highly mutated signature genes, including TP53, PIK3CA, GPR124, HRAS, SETD2, ERBB2, PNN, ATXN1, and CP (**Fig. 4a**). A few of these alterations were well-known cancer-driving events. For example, the TP53 gene, a tumor suppressor gene enabling tumors to be more aggressive, is typically mutated at the early stage of cancer to confer genetic instability to cancer cells and is associated with poor prognosis [66]. We observed that the TP53 gene was highly mutated in Basal and Her2 subtypes (**pattern ① in Fig. 4a**) and less mutated in Luminal-like subtypes. This suggested that patients with Basal and Her2 subtypes may have a worse prognosis, as proven by many studies [1, 6, 7]. PIK3CA mutation was identified as (H,LA,LB) marker, which is most common in Luminal-like breast cancer, followed by Her2 breast cancer and triple-negative breast cancer (**pattern ② in Fig. 4a**), consistent with previous studies [67, 68]. We also found HRAS, SETD2, and ERBB2 mutations to be significantly associated with Her2 breast cancer, consistent with previous reports [69-71].

UGES also identified highly mutated signature genes as novel biomarkers of intrinsic subtypes, such as GPR124, PNN, ATXN1, and CP. Notably, mutations in the ATXN1 gene were newly identified by UGES to be associated with the LumB subtype. The major phenotypical difference between LumA and LumB subtypes is that LumB tumors express a high level of Her2 and the proliferation marker Ki-67. The Notch signaling pathway regulates Ki-67 [72, 73]. Studies showed that ATXN1 is a negative regulator of the Notch pathway, and ATXN1 knockdown enhanced tumor invasion by activating the Notch pathway. Therefore, a mutated ATXN1 gene may lose its inhibitor role of the Notch pathway, and the abnormal expression of the Notch pathway leads to the overexpression of Ki-67 in Her2 cancer cells [74, 75]. GPR124 and CP gene mutations were also associated with tumor progression and invasion in breast cancer previously [76,77]. However, this is different with specific intrinsic subtypes. While our results suggest that these gene mutations are potential biomarkers of their specific intrinsic subtypes, more evidence is required to prove their functional roles.

Second, CNA alterations were found effective in identifying Basal and Her2 subtypes. As shown in **Fig. 4b** and **Table. 3**, Basal, Her2, LumA, and LumB subtypes have distinct CNA patterns. For example, PGAP3 and MIR4728 gene copy numbers were highly amplified in the Her2 subtype (**pattern ③ in Fig. 4b**), as compared to the Basal subtype. This is consistent with previous studies. For example, PGAP3 might impact on the Her2 subtype by being co-amplified with Her2 [78]; MIR4728 gene is located within an intron of the ERBB2 gene, a well-known oncogene strongly associated with the Her2 subtype [79]. Notably, UGES newly identified MIR5684 copy number deletion as a novel biomarker of the Her2 subtype, while more experimental evidence is required to confirm the relation.

Finally, in addition to genetic differences, epigenetic alterations, such as hyper- or hypo-methylation changes, also

exhibited subtype-specific patterns (**Fig. 4c**). For example, as shown in **pattern ④ of Fig. 4c**, nine genes (ADCY4, CRYAB, DNMT3, HOXA11, INS, KCNH8, LAMB3, PRTN3, UCN) showed hypomethylation in the Basal subtype. Hypomethylation generally elevates a gene's expression, and many of those genes were shown to be highly expressed in triple-negative breast cancer cells (TNBC), accounting for >80% of the Basal subtype. For example, CRYAB was known to enhance cell migration in TNBC cells, specifically [80]; and LAMB3 was known to mediate the apoptotic, proliferative, invasive, and metastatic abilities of Basal breast cancer cells [81]. Notably, UGES newly found AIRE and PRSS41 are hypermethylated in the Basal subtype, which are potential biomarkers requiring further experimental evidence.

IV. DISCUSSION

In this study, we constructed a DNA-level classifier UGES using only DNA alterations to identify breast cancer intrinsic subtypes. Through the development, we have gained several insights. First, a multi-step strategy with the grouping of LumA and LumB subtypes is key to the better performance of all classifiers. As the test set result showed (**Fig. 2c**), the best AUC scores for single and overall subtypes were all obtained from such multi-step strategies. (B,(H,(LA,LB))) had the best overall AUC scores, while (B,H,(LA,LB)) obtained the highest AUC score, 0.992, for identifying the Basal case. The AUC scores of the one-step (B,H,LA,LB) strategy were, however, consistently the lowest, in particular having trouble resolving the LumA and LumB subtypes, with AUC=0.852 and 0.782, respectively. One potential explanation is that the one-step classifier may need more accuracy to ignore the natural hierarchy of intrinsic subtypes. In fact, in the past, breast cancers have been broadly classified based on their gene expression profiles into Luminal- and Basal-type tumors. It is only recently that the Luminal type was further divided into two sub-groups [57]. This demonstrates the fact that Luminal-like subtypes differ more from Basal and Her2 subtypes than each other. Thus, we conclude that the hierarchical stepwise classification method is a more realistic approach to identifying intrinsic subtypes. The inaccuracy of the PAM50 classifier is perhaps also attributable to the fact that it took a four-classification, one-step approach.

Second, all types of DNA alterations are informative for defining subtypes. As shown in **Fig. 2d** and **Supplementary Fig. S2**, the overall AUC scores of single-feature classifiers were all over 0.6 in testing for the four strategies. Specifically, the average AUC scores of 0.698, 0.854, and 0.919 for mutation, CNA, and methylation features, respectively. It suggested that the features are all informative. Notably, the AUC scores of methylation-only classifiers were comparable to the all-feature classifiers, with the lowest score being 0.879 in the one-step multi-classification strategy. In fact, methylation changes are becoming widely accepted as biomarkers for early breast cancer detection [82]. Biologically, promoter hypermethylation is a more frequent event than

mutations in the process of carcinogenesis [83], with estimates varying from 600 to 1000 aberrantly methylated genes per tumor [84]. This significant contribution of variability may explain the strong effect of methylation in defining intrinsic subtypes and its high relative importance, as observed.

Even though methylation has a dominant influence on breast cancer subtyping, mutations and copy number alterations (CNAs) still contribute significantly to certain subtypes. For instance, somatic mutations in genes such as **TP53** and **PIK3CA** are crucial in the Basal-like and Her2-positive subtypes, serving as key driver alterations. Similarly, CNAs play a pivotal role in these subtypes, with **PGAP3** copy number gain showing high relevance in Her2-positive cases, indicating that specific chromosomal changes are important in defining these subtypes. While methylation, particularly in promoter regions, is a key marker for early detection and subtype classification, especially in Luminal A and Luminal B subtypes, more than relying on methylation data may be required [85-87]. Our findings suggest that integrating mutations, CNAs, and methylation provides a more comprehensive approach, improving the model's accuracy and generalization ability while reducing the risk of overfitting that can occur when using a single feature type.

For example, the methylation-only (B,H,LA,LB) classifier performed better in identifying the Her2 subtype (AUC = 0.943) compared to the all-feature classifier (AUC = 0.935). However, if we looked into their specific training and testing performance, the overall AUC scores for the training and test sets of the methylation-only (B,H,LA,LB) classifier were 1 and 0.879, while these for the all-feature (B,H,LA,LB) classifier, were 0.965 and 0.889, as shown in **Supplementary Fig. S2c**. The larger difference in training and test set performance of the methylation-only classifier demonstrated that it may be overfitted. Thus, only using methylation features to analyze breast cancer intrinsic subtypes would result in the loss of model generalizability.

Third, our findings provide a foundation for the development of more specific and mutually exclusive DNA biomarkers for distinguishing breast cancer subtypes. For instance, in our dataset, among the 716 samples with **TP53** mutations, only 199 also had **PIK3CA** mutations; similarly, of the 824 samples with **PIK3CA** mutations, only 199 had **TP53** mutations. This reveals a significant mutual exclusivity between **PIK3CA** and **TP53** mutations in breast cancer, a pattern consistent with previous research [88]. Such mutually exclusive mutation patterns highlight the potential to use these alterations as specific biomarkers to refine subtype classification and enhance targeted therapeutic strategies.

Fourth, the 52 DNA biomarkers we identified hold significant clinical potential. Notably, the **ATXN1** mutation shows potential specificity for the LumB subtype. Since LumA and LumB subtypes differ in treatment strategies and prognosis, **ATXN1** could serve as a valuable marker in early screening. Additionally, the association between the **MIR5684** copy number loss and the Her2 subtype suggests that **MIR5684** could enhance early screening for Her2-positive patients. Furthermore, hypermethylation of the **AIRE** and

PRSS41 genes may serve as biomarkers for the Basal subtype detection.

Beyond the signature alterations in well-known genes, many of our newly identified biomarkers have the potential to become emerging targets in breast cancer precision medicine. Some lesser-known but promising targets include **GPR124**, which influences tumor growth and metastasis in triple-negative breast cancer (TNBC) by regulating the Wnt signaling pathway. Developing small molecule inhibitors or antibodies against **GPR124** could be a potential avenue for future research. Additionally, since **ATXN1** is a negative regulator of the Notch signaling pathway, targeting **ATXN1** mutations or inhibiting the Notch pathway could provide new treatment strategies for LumB patients, particularly given their poor response to chemotherapy. The discovery of these biomarkers lays a good foundation for further research.

Another potential application for UGES is in facilitating the development of early subtyping using ctDNA liquid biopsy. Although promising, ctDNA-based breast cancer screening faces several technical challenges. First, cancer heterogeneity presents a challenge because molecular subtyping of breast cancer involves a complex interplay of genes and mutations. Since ctDNA data can only capture a portion of this information, it may not fully represent the heterogeneity of the tumor, limiting the accuracy of models trained on such data. Second, low tumor burden in the early stages of cancer results in low ctDNA concentration and mutation frequencies, making it difficult to distinguish genuine cancer-related mutations from background noise. Third, there needs to be more sufficient ctDNA datasets, and the generalizability of trained models could be better. High-quality labeled data, such as whole-genome methylation, mutation, or copy number ctDNA data, is difficult to obtain due to the scarcity of early-stage cancer samples. These limitations affect the accuracy of detection.

The insights we gained from building the UGES model can help address these challenges. First, the data used to train UGES comes from solid tumor samples, which provide higher DNA yield and quality, allowing for the more comprehensive capture of intra-tumor epi/genetic heterogeneity. Multi-patient populations, such as those represented by TCGA and METABRIC, enhance the capture of inter-tumor epi/genetic heterogeneity. Second, the high tumor burden and quality of these tumor samples ensure lower background noise, which minimizes false positives and false negatives during classifier training. Third, we carefully integrated data into a large omics dataset with 50,831 features, including mutations, copy number aberrations, and methylation patterns, from 2,065 samples. This greatly enriches the training data and improves our model's ability to manage tumor heterogeneity across subtypes. Lastly, we employed a novel multi-step hierarchical learning algorithm, rather than a traditional single-step multiclass classification approach. This strategy enables the model to learn the subtle differences between subtypes at each step, significantly enhancing its generalizability while identifying subtype-specific biomarkers. These biomarkers could potentially be used for early breast cancer screening.

Finally, our study does have several limitations worth noting. We used PAM50 subtypes as the initial training labels to construct the UGES classifier despite the fact that accurately determining the true intrinsic subtypes of breast cancer experimentally is challenging [4]. We opted for the PAM50 label because it is one of the most internationally recognized for classifying intrinsic subtypes. However, unlike the typical approach that uses expression data, UGES was developed using only DNA alterations. This method yielded highly accurate results and produced clinically more distinguishable subtype groups, demonstrating that DNA alterations can serve as sufficient and reliable features. Nevertheless, UGES may not capture some emerging DNA mutations or features. While our current sample size is adequate to distinguish the major subtypes, smaller subtypes have been difficult to identify due to insufficient data. As new subtype discoveries emerge [89-91] and more data become available, future models should incorporate these refined subtypes and additional feature information. Also, we have not yet considered chromosomal-level structural changes [92, 93] and mid-size insertion-deletions, which challenge detection by next-generation sequencing-based technologies [94] but are important to understanding breast cancers. Potential bias from confounding clinical factors, such as tumor size and lymph node metastasis, could influence survival outcomes and impact the interpretation of subtypes and survival. Future analyses should consider additional clinical variables, such as tumor size and treatment modalities, to further adjust for these confounders.

DATA AVAILABILITY STATEMENT

The TCGA and METABRIC datasets are available in GDC (<https://gdc.cancer.gov/about-data/publications/pancanatlas>) and cBioPortal (<http://www.cbioportal.org/>). Analytical codes are provided at <https://github.com/labxscut/UGES>.

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Fig. 1. The overall study design, analysis workflow, and dataset description: (a) The principles of DNA-level subtype classifier; (b) Distribution of the four intrinsic subtypes in the TCGA ($n = 931$) and METABRIC ($n = 1134$) breast cancer datasets (c) The conceptual design of UGES.

Fig. 2. Evaluation of the classification strategies on DNA-level features: (a) The self-learned classification hierarchy (B,(H,(LA,LB))); (b) Competitive classification hierarchies: ((B,H),(LA,LB)), (B,H,(LA,LB)), and (B,H,LA,LB); ROC and AUC of (c) all strategies with full DNA features; (d) the (B,(H,(LA,LB))) strategy with mutation, CNA, and methylation features.

Fig. 3. UGES subtypes show improved clinical relevance: (a) Alluvial plot and (b) Confusion matrix between UGES and PAM50 subtyping for the combined TCGA+METABRIC cohort; (c) Changes of p-values when switching from PAM50 to UGES subtypes; (d) Forest plot of multivariate model hazard ratios.

Fig. 4. Heatmap of the 52 signature subtype-delineating alterations: from top to bottom are heatmaps of signature (a) Mutations, (b) Copy Number Aberrations, and (c) Methylation features.